

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
.Net Core and Progressive Web Application

Practical – IX

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Class: TYIT	Batch: 1
Date of Assignment: 18/08/2026	Date/Time of Submission: 18/08/2026

A) Develop CRUD operations using ASP.NET Core Web API and Entity Framework Core.

ALGORITHM:

Install below Packages:

1. Microsoft.EntityFrameworkCore
2. Microsoft.EntityFrameworkCore.SqlServer
3. Microsoft.EntityFrameworkCore.Tools
4. Swashbuckle.AspNetCore
5. Microsoft.EntityFrameworkCore.Design

CODE:

StudentController.cs:-

```
using Microsoft.AspNetCore.Mvc;
using WebApplication17.Data;
using WebApplication17.Models;
namespace WebApplication17.Controllers
{
    [Route("api/[controller]")]
    [ApiController]
    public class StudentsController : ControllerBase
    {
        private readonly AppDbContext _context;
        public StudentsController(AppDbContext context)
        {
            _context = context;
        }
        // GET ALL
        [HttpGet]
        public IActionResult GetStudents()
        {
```

```
        return Ok(_context.Students.ToList());
    }
    // GET BY ID
    [HttpGet("{id}")]
    public IActionResult GetStudent(int id)
    {
        var student = _context.Students.Find(id);
        if (student == null)
            return NotFound();
        return Ok(student);
    }
    // INSERT
    [HttpPost]
    public IActionResult AddStudent(Student student)
    {
        _context.Students.Add(student);
        _context.SaveChanges();
        return Ok(student);
    }
    // UPDATE
    [HttpPut("{id}")]
    public IActionResult UpdateStudent(int id, Student student)
    {
        var data = _context.Students.Find(id);
        if (data == null)
            return NotFound();
        data.Name = student.Name;
        data.Age = student.Age;
        data.Course = student.Course;
        _context.SaveChanges();
        return Ok(data);
    }
    // DELETE
    [HttpDelete("{id}")]
    public IActionResult DeleteStudent(int id)
    {
        var student = _context.Students.Find(id);
        if (student == null)
            return NotFound();
        _context.Students.Remove(student);
        _context.SaveChanges();
        return Ok(student);
    }
}
```

```
}  
}
```

AppDbContext:-

```
using Microsoft.EntityFrameworkCore;  
using WebApplication17.Models;  
namespace WebApplication17.Data  
{  
    public class AppDbContext:DbContext  
    {  
        public AppDbContext(DbContextOptions<AppDbContext> options) : base(options) { }  
        public DbSet<Student> Students { get; set; }  
    }  
}
```

Student.cs:-

```
namespace WebApplication17.Models  
{  
    public class Student  
    {  
        public int Id { get; set; }  
        public String Name { get; set; }  
        public int Age { get; set; }  
        public String Course { get; set; }  
    }  
}
```

Program.cs:-

```
using Microsoft.EntityFrameworkCore;  
using WebApplication17.Data;  
var builder = WebApplication.CreateBuilder(args);  
builder.Services.AddControllers();  
builder.Services.AddDbContext<AppDbContext>(options =>  
    options.UseSqlServer(  
        builder.Configuration.GetConnectionString("DefaultConnection")));  
builder.Services.AddEndpointsApiExplorer();  
builder.Services.AddSwaggerGen();  
var app = builder.Build();  
app.UseSwagger();  
app.UseSwaggerUI();  
app.UseHttpsRedirection();  
app.UseAuthorization();  
app.MapControllers();
```

```
app.Run();
```

OUTPUT:

The screenshot displays a REST client interface with two tabs. The top tab, titled 'POST /api/Students', shows the request configuration. The 'Parameters' section is empty. The 'Request body' is set to 'application/json' and contains a JSON object:

```
{  "id": 0,  "name": "Ritesh",  "age": 20,  "course": "BscIT"}
```

. The bottom tab, titled 'GET /api/Students/{id}', shows the response details for a GET request. The 'Request URL' is 'https://localhost:44328/api/Students'. The 'Server response' shows a status code of 200. The 'Response body' contains a JSON object:

```
{  "id": 4,  "name": "Ritesh",  "age": 20,  "course": "BscIT"}
```

. The 'Response headers' section lists:

```
content-type: application/json; charset=utf-8  date: Tue, 18 Aug 2026 03:31:33 GMT  server: Microsoft-IIS/10.0  x-powered-by: ASP.NET
```

POST /api/Students

Parameters Cancel Reset

No parameters

Request body application/json

Edit Value | Schema

```
{  "id": 0,  "name": "Ritesh",  "age": 20,  "course": "BscIT"}
```

Curl

```
curl -X 'POST' \  'https://localhost:44328/api/Students' \  -H 'accept: */*' \  -H 'Content-Type: application/json' \  -d '{  "id": 0,  "name": "Ritesh",  "age": 20,  "course": "BscIT"  }'
```

Request URL

https://localhost:44328/api/Students

Server response

Code Details

200

Response body

```
{  "id": 4,  "name": "Ritesh",  "age": 20,  "course": "BscIT"}
```

Response headers

```
content-type: application/json; charset=utf-8  date: Tue, 18 Aug 2026 03:31:33 GMT  server: Microsoft-IIS/10.0  x-powered-by: ASP.NET
```

GET /api/Students/{id}

Parameters Cancel

Name	Description
id * required integer(\$int32) (path)	1

Responses

Curl

```
curl -X 'GET' \
  'https://localhost:44328/api/Students/1' \
  -H 'accept: */*' \
  -H 'Content-Type: application/json'
```

Request URL

https://localhost:44328/api/Students/1

Server response

Code	Details
200	Response body <pre>{ "id": 1, "name": "Ritesh", "age": 20, "course": "BscIT" }</pre> Response headers <pre>content-type: application/json; charset=utf-8 date: Tue, 18 Aug 2026 03:31:06 GMT server: Microsoft-IIS/10.0 x-powered-by: ASP.NET</pre>

PUT /api/Students/{id}

Parameters

Cancel Reset

Name	Description
id * required	
integer(int32)	4
(path)	

Request body

application/json

Edit Value | Schema

```
{
  "id": 4,
  "name": "Rakesh",
  "age": 20,
  "course": "BscIT"
}
```

Curl

```
curl -X 'PUT' \
  'https://localhost:44328/api/Students/4' \
  -H 'accept: */*' \
  -H 'Content-Type: application/json' \
  -d '{
    "id": 4,
    "name": "Rakesh",
    "age": 20,
    "course": "BscIT"
  }'
```

Request URL

https://localhost:44328/api/Students/4

Server response

Code	Details
200	Response body <pre>{ "id": 4, "name": "Rakesh", "age": 20, "course": "BscIT" }</pre> Response headers <pre>content-type: application/json; charset=utf-8 date: Tue, 18 Aug 2026 03:34:15 GMT server: Microsoft-IIS/10.0 x-powered-by: ASP.NET</pre>

Responses

Code	Description	Links
200	OK	No links

DELETE /api/Students/{id}

Parameters Cancel

Name	Description
id * required integer (\$int32) (path)	4

Execute Clear

Responses

Curl

```
curl -X 'DELETE' \
'https://localhost:44328/api/Students/4' \
-H 'accept: */*'
```

Request URL

```
https://localhost:44328/api/Students/4
```

Process: [15944] iisexpress.exe Lifecycle Events Thread: [21284] .NET TP Worker Stack Frame: [Except

SQL Server Object Explorer

(localdb)\MSSQLLocalDB (SQL Server 17.0.4025 - I

Databases

System Databases

EmployeeDB

StudentDB

Tables

dbo.Students [Data] SQLQuery2.sql dbo.Students [Design]

Max Rows: 1000

	Id	Name	Age	Course
	1	Ritesh	20	BscIT
	2	Saqlain	20	BscIT
	3	Himani	20	BscIT
	NULL	NULL	NULL	NULL

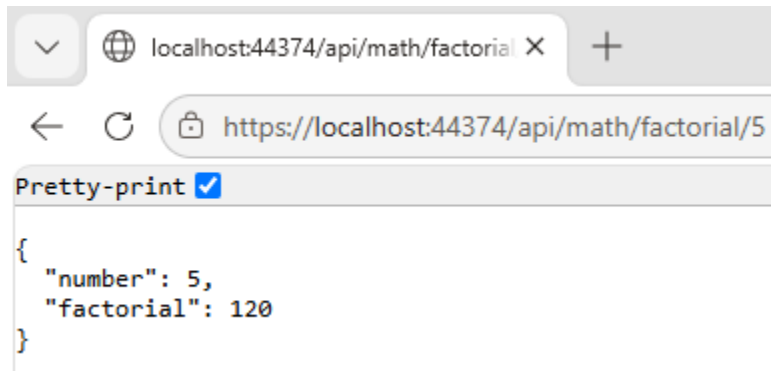
B) Create an API endpoint to calculate Factorial of a Number and test it using Postman.

CODE:

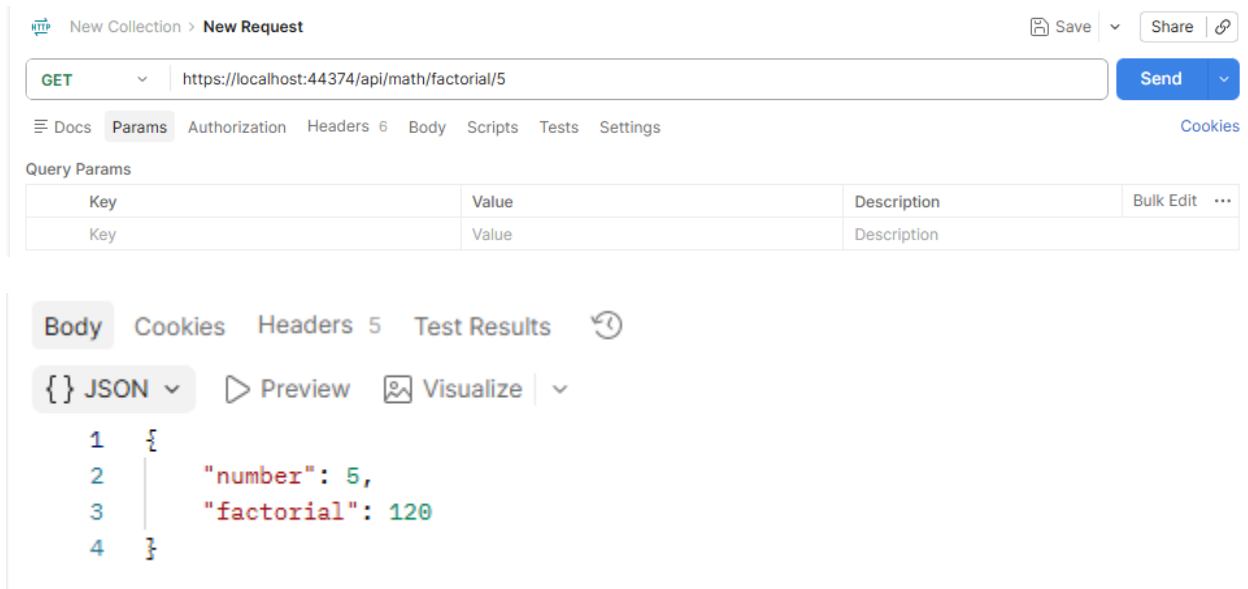
```
using Microsoft.AspNetCore.Mvc;
namespace WebApplication16.Controllers
{
    [Route("api/[controller]")]
    [ApiController]
    public class MathController : ControllerBase
    {
        [HttpGet("factorial/{number}")]
        public IActionResult GetFactorial(int number)
        {
            if (number < 0)
            {
                return BadRequest("Factorial not possible for -ve nos.");
            }
            long factorial = 1;
            for (int i = 1; i <= number; i++)
            {
                factorial *= i;
            }
        }
    }
}
```

```
    }  
    return Ok(new  
    {  
        Number = number,  
        Factorial = factorial  
    });  
    }  
}  
}
```

OUTPUT:



Verification with Postman:



HTTP

New Collection > New Request

Save

Share

GET

https://localhost:44374/api/math/factorial/-2

Send

Docs

Params

Authorization

Headers 6

Body

Scripts

Tests

Settings

Cookies

Query Params

Key	Value	Description	Bulk Edit	...
Key	Value	Description		

Body

Cookies

Headers 5

Test Results

Raw

Preview

Debug with AI

1

Factorial not possible for -ve nos.